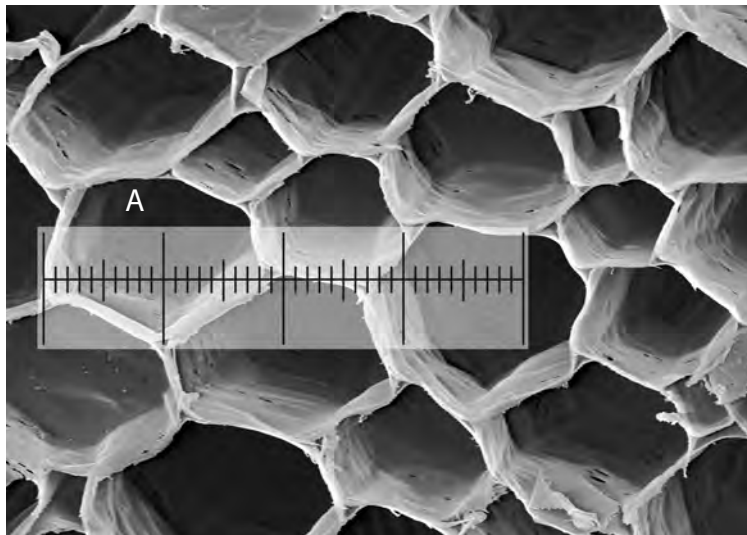


- 2 Fibres from the stems of hemp plants are widely used in industry for their great tensile strength.

The relationship between tensile strength and fibre diameter was investigated.

The photograph shows a cross section of a part of a hemp fibre and an eyepiece graticule. The fibre bundle is made up of pith cells, some of which are shown here.



(Source: © Ted Kinsman/SCIENCE PHOTO LIBRARY)

- (a) (i) Each of the smallest units on the graticule is 3×10^{-6} m.

Calculate the width of the pith cell labeled A in micrometres (μm).

(2)

Answer μm

- (ii) Name the **two** types of tissue found in fibres from plant stems.

(2)

- 1
- 2



-
- Diagram illustrating a simple pendulum setup. A horizontal line represents the support. Two vertical lines, labeled "stand", are attached to the ends of this support. A single line, labeled "hemp fibre", is attached to the center of the support and hangs down to a horizontal bar. A label "50 g mass" points to this bar.

- (4)

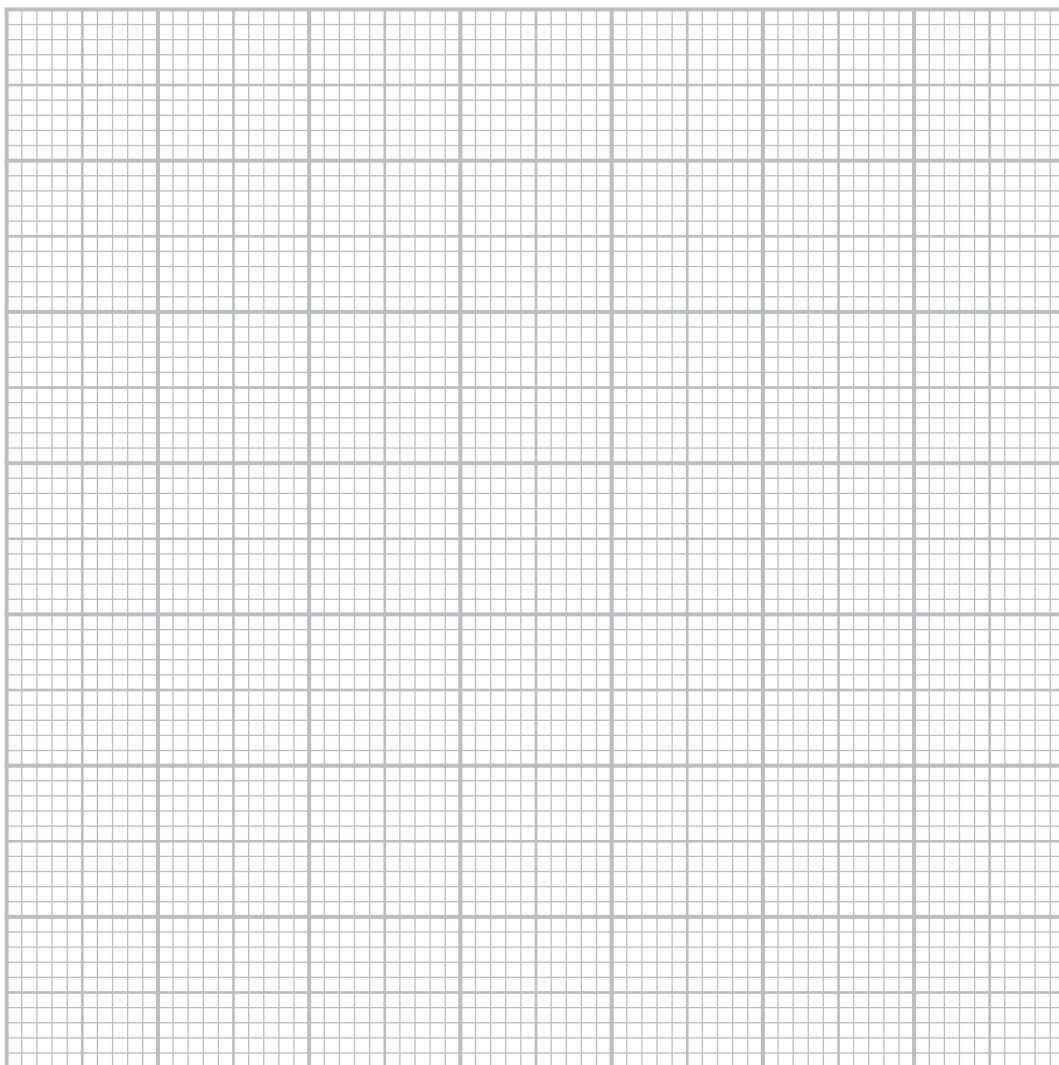
[illegible]

(ii) Some results from this investigation are shown in the table.

Diameter / μm	Tensile strength / MPa
21	1280
25	830
40	610
43	580
60	390
81	190
84	180

Plot a suitable graph of these results. Join the points with straight lines.

(4)



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(iii) Describe the relationship between the diameter and the tensile strength of these fibres.

(3)

(Total for Question 2 = 15 marks)

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